Lecture 4 Dictionary Problem and its solution using Hashing

31 July 2026 11:06

Motivating example:

- Bank account scenario:
  - Each account is identified by an account number
  - Each account stores additional information such as
    - Balance,
    - Name & address of account holder
    - History of transactions
- Any application that wish to operate on account need to provide the account no as key

The dictionary:

- Dictionaries is a data structure that store elements so that they can be located quickly using keys
- It provides and abstract model of database such that
  - Stores (key, element) pair
  - The main operations supported by dictionary are
    - FindElement (key),
    - insertItem(key, Element),
    - deleteItem(Key),

Following data structure can be used to implement dictionary:

- Array & linked list
- Hash tables
- Binary tree
- Red black tree
- AVL tree
- B tree

Searching problem:

Input: Sequence of numbers (database)
A single number (query)

Output: Index of the number, if found or NIL

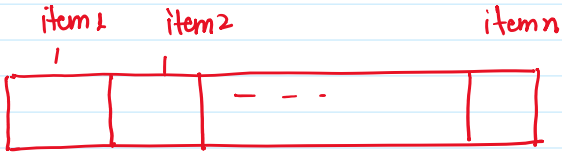
int val1

int val2

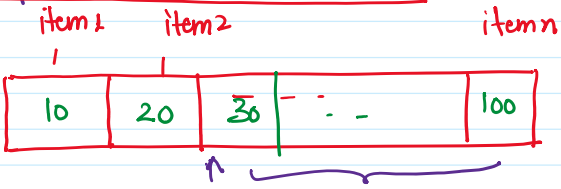
⋮

}

Create array of struct



① Generate array of struct in sorted order



ⓐ Search O(logn) Binary Search

ⓑ Insert - (key, val) := item First need to perform search to find appropriate position O(logn)

Shift O(n)

Overall = O(n)

ⓒ Delete (key) First need to perform search to find appropriate position O(logn)
Shift O(n)
} Overall = O(n)

Approach is useful when there are frequent search but rare insert and delete.

② Store elements in un-ordered / (unsorted) sequence

ⓐ Search O(n) linear Search

ⓑ Insert (key, val) O(1) := insert new element at last

ⓒ Delete (key) search — O(n)
shift — O(n)
} Overall O(n)

Approach is useful when there are frequent insert but rare search and delete.

Consider bank account scenario. Let acc no. consist of 10 digits

of account no. possible 10^{10}

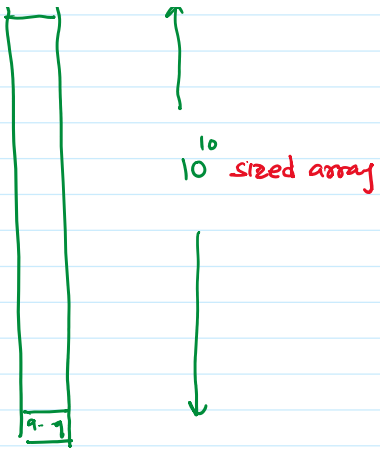


Search O(1)

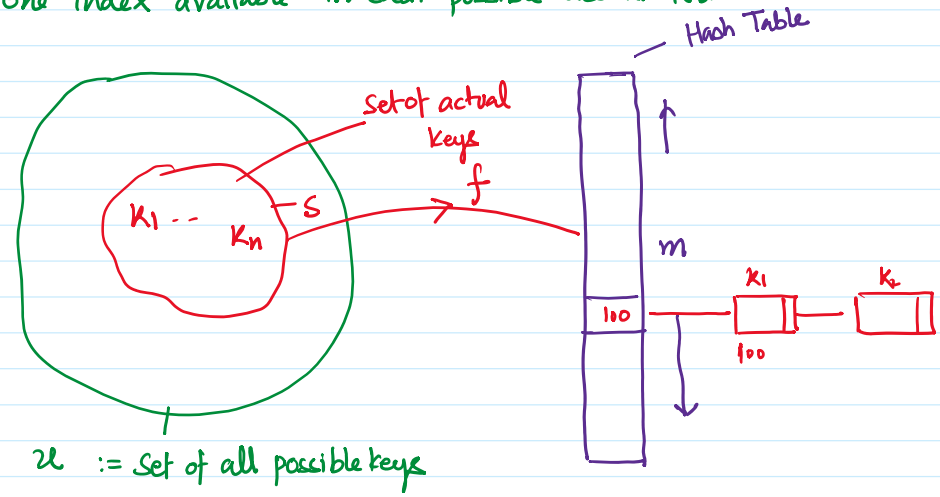
Search O(1)

Insert O(1)

Delete O(1)



There is one index available for each possible account No



$|U| \gg |S| = n$ $\quad m < n$ $\quad f(k_1) = f(k_2)$

Need a fn $f: S \longrightarrow \{0, \ldots m-1\}$ (key, val)

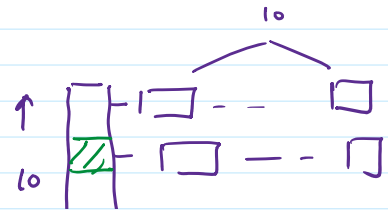
↓ Input as keys ↓ index location in array

Since $m<n$, more than two keys can map into same index := collision

Address collision by chaining

⇒ One candidate for function f is $f(k) = k \bmod m$ $\quad m=5$

5, 10, 15, 20, 25 . . .

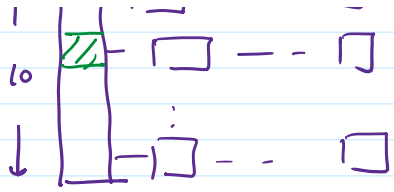


As $m<n$ $\quad n = 100$ $\quad m = 10$

if $m = \sqrt{n}$

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if $m = \sqrt{n}$

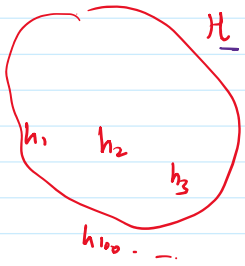


Aim is to find map f that uniformly distribute all keys into m slot.

Universal Hashing : Definition

Universe $= \mathcal{U}$ $\quad$ m

$S = \{k_1 \ldots k_n\} \quad S \subset \mathcal{U}$



A family of hash fn H.

- Choose a hash function $h \in H$ uniformly at random
- We call H as universal hash family if $\forall$ pair of keys $k \neq k' \in S$

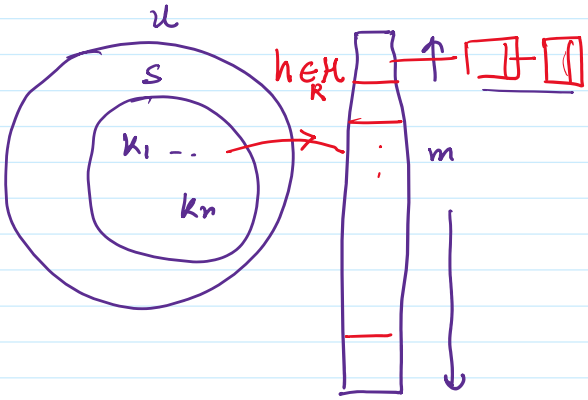
$$\Pr_{h \in H}[h(k) = h(k')] \leq \frac{1}{m} \quad -(1)$$

$h \in H$ universal hash fn.

Th Let H be a universal hash family. Let $S = \{k_1 \ldots k_n\}$ be set of actual keys

$$E[\#\text{ keys colliding in same slot}] \leq \frac{n}{m}.$$

$n > m$



$m = \sqrt{n}$.

Proof Consider $k_1 \ldots k_n$ (keys)

$E[\#\text{ keys colliding with } k_i] \quad \forall i \in \{1 \ldots n\}$

$$I_{ij} = \begin{cases} 1 & \text{if } h(k_i) = h(k_j) \quad i \neq j \\ 0 & \text{o.w} \end{cases}$$

Indicator r.v. $i, j \in \{1 \ldots n\}$

$$E[\#\text{ keys colliding with } k_i] = E\left[\sum_{i \neq j} I_{ij}\right] = \sum_{i \neq j} E[I_{ij}]$$

$i, j \in \{1 \ldots n\}$

$$= \sum_{i \neq j} \left[1 \cdot \Pr[I_{ij} = 1] + 0 \cdot \underbrace{\Pr[I_{ij} = 0]}_{0}\right]$$

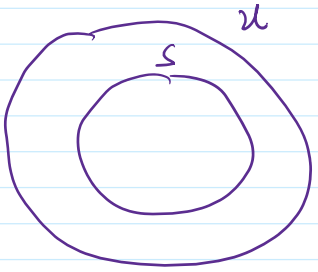
$$= \sum_{i \neq j} \Pr[h(k_i) = h(k_j)]$$

$$\leq \sum_{i \neq j} \frac{1}{m} = \frac{n-1}{m}$$

Bad example of Universal Hash fn

$\mathcal{U}$

Bad example of Universal Hash fn



$S \subset U$

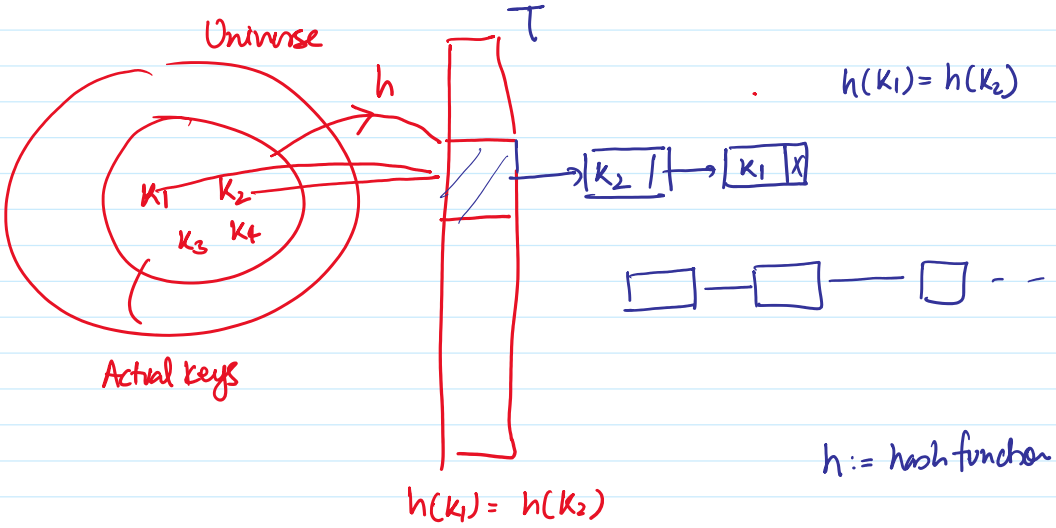
$H = \{ h : \{0 \ldots |U|-1\} \longrightarrow \{0 \ldots m-1\} \}$

For each possible element of U generate a random no. from $\{0 \ldots m-1\}$

Space complexity = $O(|U| \log m)$

↑ very large

$k_1 \in S$ (set of actual keys)



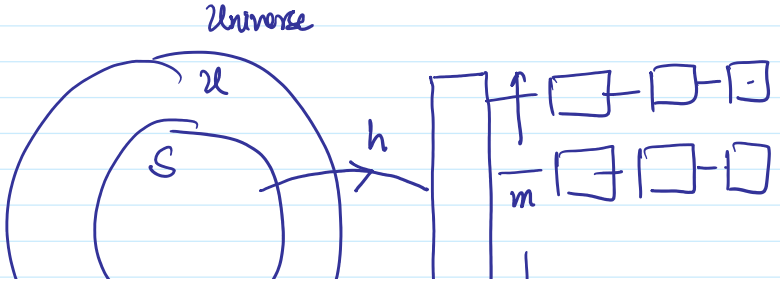
h := hash function

U : Set of all possible acc no. of 11 digit

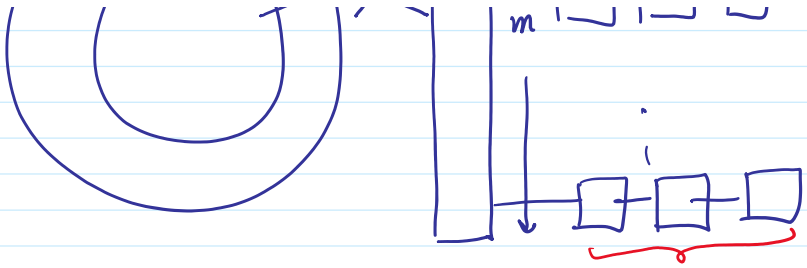
$m = 5000$

$h(k) = k \bmod m \qquad k \in U$

$\downarrow m$



$|S| = n$



Dot product hash family

base 3

- assume m is prime
- Size of universe $|U| = m^r$
- view key K in base m representation $K := \langle k_0, k_1, k_2 \ldots k_{r-1} \rangle$

  $0 \leq k_i \leq m-1 \quad \forall i \in \{0 \ldots r-1\}$
- For key $a = \langle a_0, a_1 \ldots, a_{r-1} \rangle \quad 0 \leq a_i \leq m-1$

$$h_a(K) := \left(\sum_{i=0}^{r-1} a_i k_i \right) \bmod m$$

Dot product Hash family $\mathcal{H} = \{ h_a \mid a \in \{0 \ldots |U|-1\} \}$

Space complexity $O(\log |U|)$

$m = 5 \quad r = 3$

$|U| = 125$

$K = 64 = \langle 2\ 2\ 4 \rangle$

$a = 96 = \langle 3\ 4\ 1 \rangle$

$h_a(K) = (6 + 8 + 4) \bmod 5$

$= 3$

Th Dot product hash family is universal.

Proof: We call $\mathcal{H}$ as universal hash family if $\forall$ pair of keys $K \neq K' \in S$

$$\Pr_{h \in \mathcal{H}} [h(K) = h(K')] \leq \frac{1}{m} \quad \text{--- (1)}$$

Consider two keys $K \neq K'$ they differ on atleast one digit say d-th

$K = \langle k_0, k_1 \ldots k_d, \ldots k_{r-1} \rangle$

$K' = \langle k_0', k_1', \ldots k_d' \ldots k_{r-1}' \rangle$

Since $K \neq K'$, w.l.o.g. Assume that $k_d \neq k_d'$

Since $k \neq k'$, w.l.o.g. Assume that $k_d \neq k'_d$

$$\Pr_{h_a \in H}\left[h_a(k) = h_a(k')\right] = \Pr_{h_a \in H}\left[\sum_{i=0}^{r-1} a_i k_i \bmod m = \sum_{i=0}^{r-1} a_i k'_i \bmod m\right]$$

$$= \Pr\left[\sum_{i=0}^{r-1} a_i (k_i - k'_i) \equiv 0 \bmod m\right]$$

$$= \Pr\left[a_d (k_d - k'_d) + \sum_{i \neq d} a_i (k_i - k'_i) \equiv 0 \bmod m\right]$$

$$= \Pr\left[a_d = -(k_d - k'_d)^{-1} \sum_{i \neq d} a_i (k_i - k'_i) \bmod m\right]$$

↑ inverse of $(k_d - k'_d)$ in multiplicative group mod m

Multiplicative group modulo p

$p = 7 = \{1, 2, 3, 4, 5, 6\}$

$2 \times 3 = 6 \pmod 7$

$3 \times 5 = 1 \pmod 7$ $\quad 3^{-1} = 5$

$a\,b = 1 \pmod b$

$a \Leftrightarrow b^{-1}$

$f(k, k', \{a_0 \ldots a_{r-1}\}/\{a_d\})$

$$= \Pr\left[\underbrace{a_d}_{x} = \underbrace{-(k_d - k'_d)^{-1}}_{a} \underbrace{\sum_{i \neq d} a_i (k_i - k'_i)}_{b} \bmod \underbrace{m}_{p}\right] = \frac{m^{r-1}}{m^r} = \frac{1}{m}$$

$ax \equiv b \pmod p$ has unique soln if $\gcd(a, p) = 1$

Once the value of $\{a_0 \ldots a_{r-1}\}/\{a_d\}$ is selected, then exactly one value of a_d causes k and k' to collide.

Counting the possible value of a_d

$a = \langle a_0 \ldots \underset{\uparrow}{a_d} \ldots a_{r-1} \rangle$

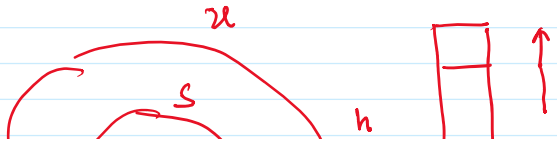
m possibilities $\{0 \ldots m-1\}$

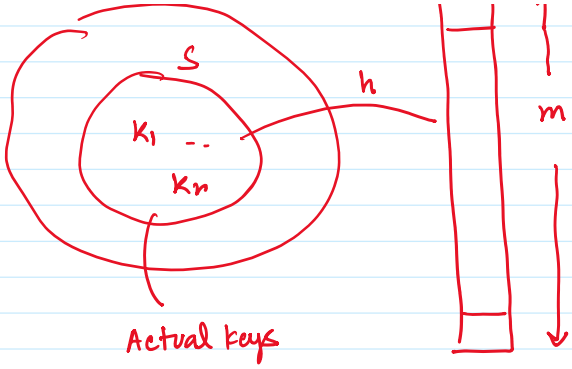
Another Universal Hash family $\quad h_{ab}(k) = ((ak + b) \bmod p) \bmod m$

$H = \{h_{ab} \mid a, b \in \{0 \ldots |u|-1\}\}$

b is a prime no. greater than m

CLRS





h universal hash family

$|\mathcal{U}| \gg |S| = n$

$$I_{ij} = \begin{cases} 1, & \text{if } h(K_i) = h(K_j) \\ 0, & \text{o.w} \end{cases}$$

$[n] = \{1 \cdots n\}$

Total no. of collision $= \sum_{\substack{i,j \in [n] \\ i \neq j}} I_{ij}$

$$E[\#\text{of collision}] = E\left[\sum_{i \neq j} I_{ij}\right] = \sum_{i \neq j} E[I_{ij}] = \sum_{i \neq j} \Pr[h(K_i) = h(K_j)]$$

$$\leq \sum_{\substack{i \neq j \\ i,j \in [n]}} \frac{1}{m}$$ since h is universal hash fn

$$= \frac{n(n-1)}{2m} \quad \text{if } m = 5n^2$$

$$< \frac{1}{10}$$

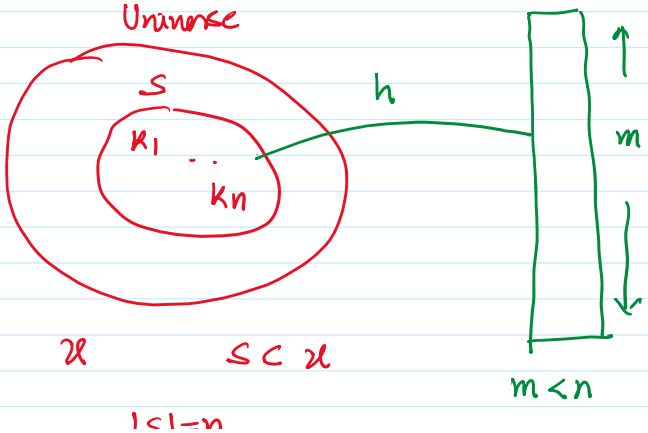
$$\Pr[\#\text{of collision} > 1] \leq \frac{E[\#\text{of collision}]}{1}$$

$$\leq \frac{1}{10}$$

Let X be a non-negative r.v

$\Pr[X > a] < \frac{E[X]}{a}$

$\Rightarrow$ w.p atleast $\frac{9}{10}$ there will not be any collision.



$h(\bar{k}) \in \{0 \cdots m-1\}$

$\mathcal{U}$ $\quad S \subset \mathcal{U}$ $\quad |S| = n$ $\quad m < n$

① when $m < n$ then there will be collision

Universal hash fn gurantees that $E_{h \in_R H}[\#\text{ of keys mapped in same slot}] < \frac{n}{m}$

① Advantage : Size of hash table is small

② Disadvantage : Search time is higher [if $m = \sqrt{n}$, then Expected search time $O(\sqrt{n})$]

② when $m = 5n^2$

Universal hash fn gurantees that w.p at 9/10 there will not be any collision

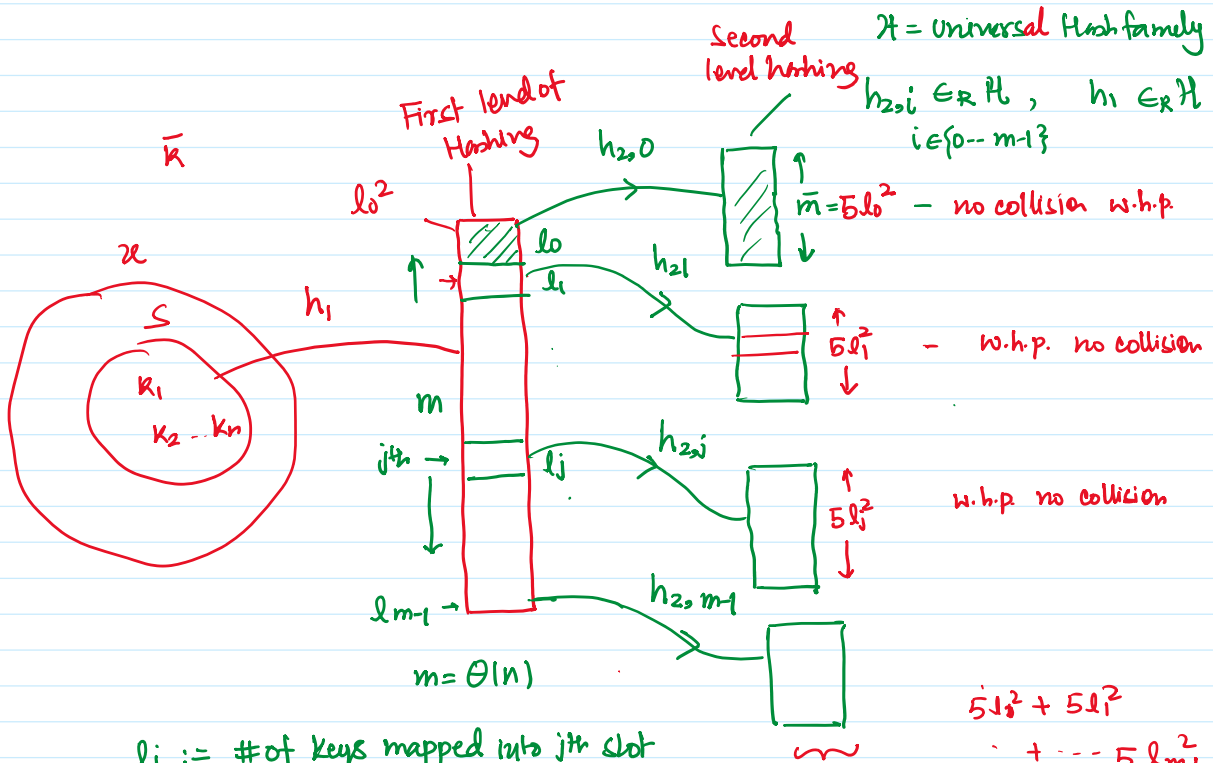
Advantage: Search time is $\Theta(1)$

Disadvantage : Size of hash table $m = O(n^2)$ become larger

Question Can we have search time $\Theta(1)$ with $m = \Theta(n)$

No. of keys are fixed. Static dictionary Problem

Perfect Hashing



l_i := # of keys mapped into j^{th} slot

$m = \Theta(n)$

l_j := # of keys mapped into j^{th} slot

$5l_0^2 + 5l_1^2 + \cdots + 5l_{m-1}^2$

$h_1(\bar{k}) \in \{0 \cdots m-1\}$

$h_1(\bar{k}) = 1 \qquad h_{2,1}(\bar{k}) \in \{0 \cdots 5l_1^2\}$

Step1 Pick $h_1 : \{0 \cdots |U|-1\} \longrightarrow \{0 \cdots m-1\}$ from Universal hash family

Step 1.5 If $\sum_{i=0}^{m-1} 5l_i^2 > cn$ then redo step1.

Step2 For each slot in first hash table $j \in \{0 \cdots m-1\}$

- l_j = # of keys mapped in j^{th} slot
- pick $h_{2,j} : \{0 \cdots |U|-1\} \to \{0 \cdots 5l_j^2 - 1\}$ from universal hash family
- while $h_{2j}(k_i) = h_{2j}(k_i')$ for some i, $k_i \neq k_i'$ with $h_1(k_i) = h_1(k_i')$
  - Repick $h_{2,j}$

// w.p. $9/10$ there will not be any collision

Claim Total size of hash table used in 2nd level hashing is small

Proof

$$E\left[\sum_{i=0}^{m-1} 5l_i^2\right] = 5\,E\left[\sum_{i=0}^{m-1} l_i^2\right] = 5E\left[\sum_i \sum_j I_{ij}\right]$$

($\sum_{i=0}^{m-1} l_i^2$ captures collision due to h_1)

$$I_{ij} = \begin{cases} 1 & \text{if } h_1(k_i) = h_1(k_j) \\ 0 & \text{o.w} \end{cases}$$

l_j^2 := captures no. of collision due to h_1 for j^{th} index

$$= 5 \sum_i \sum_j E[I_{ij}]$$

$i, j \in \{1 \cdots n\}$

$$\leq 5 \sum_i \sum_j \frac{1}{m} = \frac{5n^2}{2m}$$

as we choosed $m = \Theta(n)$

$$E\left[\sum_{i=0}^{m-1} 5l_i^2\right] < \Theta(n)$$

Total size of 2nd level hash table

$$\Pr\left[\sum_{i=0}^{m-1} 5l_i^2 > c \cdot n\right] < E\left[\sum_{i=0}^{m-1} 5l_i^2\right] \quad \Theta(n)$$

$$Pr\left[\sum_{i=0}^{m-1} 5l_i^2 > c.n\right] \leq \frac{E\left[\sum_{i=0}^{m-1} 5l_i^2\right]}{c.n} = \frac{\Theta(n)}{cn} < \frac{1}{10}$$

↑ Bad event

↑ choosing const c appropriately

Total space complexity $= \Theta(n) + \sum_{i=0}^{m-1} 5l_i \leq \Theta(n) + cn = \underline{\Theta(n)}$

↑ First level hash fn

(Fredman, Komlos, Szemeredi 1984)

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